

Introduction to Soil Science

Soil science is the study of the soil as a natural resource and this includes the study of various components of the soil such as the soil formation, classification, mapping, properties and management.

It also includes the study of the soils physical (texture, structure, water holding capacity), chemical (nutrient availability, soil pH) and biological (soil living organisms) properties of the soil.

This study helps to effectively conserve and manage the soil for efficient maintenance of soil fertility and nutrient availability to plants.

You can find soil scientists working in:

1. Agricultural sector: where they contribute to improving soil fertility.
2. Environmental sectors: where they contribute to ecosystem reclamation and restoration studies and practices.
3. Urban planning sectors: where they contribute to construction and building planning, effective waste management.
4. Climate change research sectors: where they study the impact of practices on soil to carbon cycle and climate change.

The whole essence of soil science is to study and effectively manage the soil.

Principles of Soil Science

Soil is the uppermost part of the earth's surface where plants grow. It is formed from weathered rock and decomposed organic matter. It consists of mineral compounds, decayed organic materials, living organisms, air, water and nutrients all interacting to support life.

Soil components

1. Mineral Matter: These are particles derived from weathered rock and minerals. They include:

- Sand: large particles, improves drainage
- Silt: medium-sized particles, holds nutrients
- Clay: small particles, holds water and nutrients strongly

2. Organic Matter: Decomposed remains of plants and animals. It:

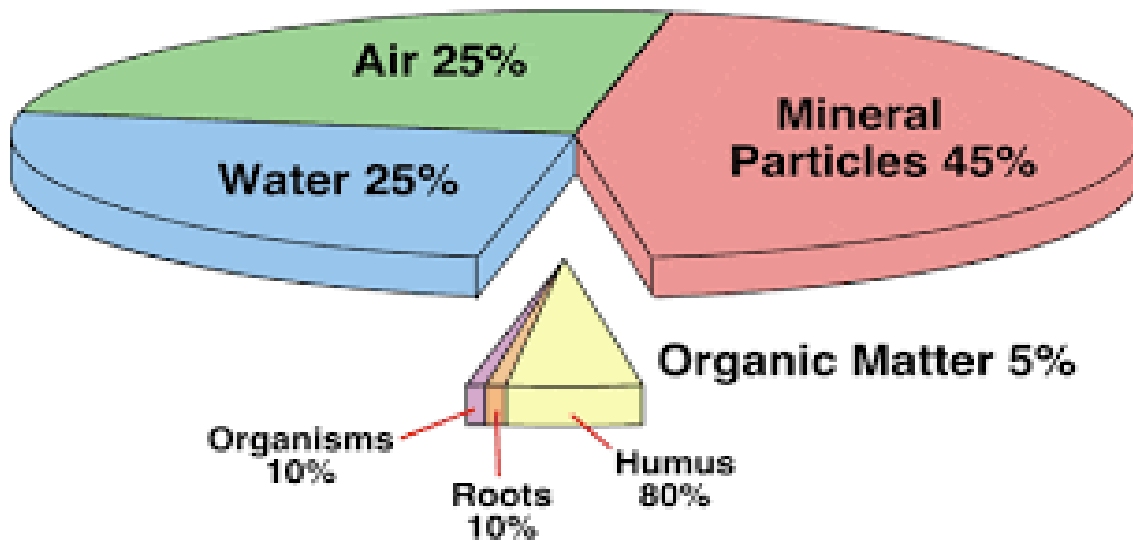
- Improves soil structure

- Increases water-holding capacity
- Provides nutrients as it decomposes

3. Water: Essential for plant growth. It dissolves nutrients for root absorption.

4. Air: Provides oxygen for plant roots and soil organisms.

5. Living Organisms: Includes earthworms, fungi, and bacteria. These help decompose organic matter, improve nutrient cycling, and structure the soil.



The minerals and organic matter are responsible for storing plant nutrient; the water component is responsible for providing nutrient for plant uptake while the air component provides the soil microorganisms with air needed for the biological processes that release nutrients in the soil. The quality of the soil is determined by adequate presence of all these components.

Soil is an important component of the Agricultural land because:

1. It is the main medium for plant growth: it contains the nutrients necessary to support plant growth and it holds the plant root in place and protects it from direct sunlight.
2. Soil provides water storage: It helps to retain moisture from rainfall and makes them available for plant use. The amount of moisture retained differs from soil to soil.
3. Soil is a source of air: this provides the oxygen required for the breakdown of sugar into energy needed by plants and soil organisms to grow.
4. Soil is a home to various macro and microorganisms.

5. Soil takes part in all major biogeochemical cycles: Nitrogen, Carbon, and Phosphorus cycles. These cycles help to maintain a balance.

Soil Formation and Classification

Soil Formation (Pedogenesis):

Soil is formed over long periods through the interaction of several factors and processes:

Factors Influencing Soil Formation:

1. Parent Material: Original rock from which the soil develops.
2. Climate: Rainfall and temperature affect weathering rates.
3. Organisms: Plants and animals contribute organic matter and mixing.
4. Topography: Slope influences erosion and soil depth.
5. Time: More time allows more development and horizon formation.

Processes in Soil Formation:

For a healthy layer of soil to be formed, the following processes take place.

Weathering: This involves the breakdown of rocks into smaller pieces through physical or chemical processes.

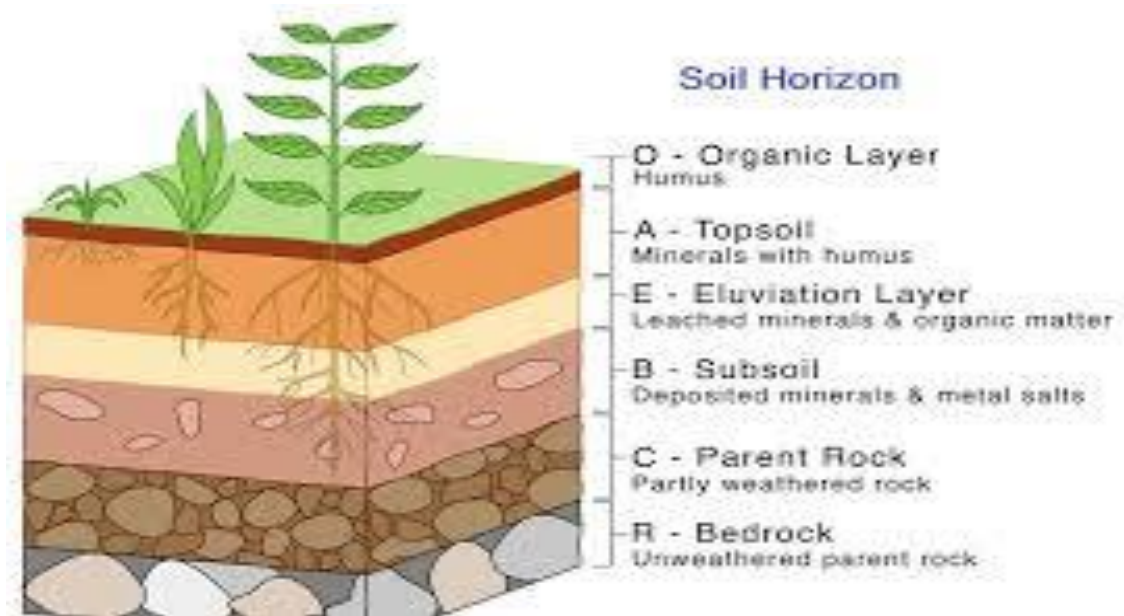
Transportation of weathered rocks or sediment erosion: the weathered rock/ sediment needs to be transported or eroded so that it can accumulate in a location where soil is to be formed.

Organic matter introduction and decomposition: This results from death of plants and animals.

1. Build-up of organic matter and continuous accumulation.

Soil profile

This is a vertical cross section of the soil that shows various layers of the soil each running parallel to the surface. These layers are known as horizons. A soil profile helps to distinguish a particular soil sample from the rest. Distinguishing factors include its colour, texture, structure, thickness or chemical composition.



The horizons from the image above are represented by letters O A E B C and R to show the activities that go on within each layer.

The O horizon: Is the uppermost part of the soil profile which is also visible to the eye. It is composed of the organic materials such as, leaves, twigs, grasses and decomposing organic matter. The presence of organic matter on this layer gives it, its striking black or dark brown colour.

The A horizon: otherwise known as the topsoil is where seed germination occurs, development of new roots that transforms into a plant also takes place here. This layer is rich in organic matter. It is soft and porous enough to hold water and nutrients. It contains microorganisms such as earthworms, bacteria and fungi.

The E horizon: This is the zone of eluviation. This implies that it contains all the materials leached/washed down from zone O and A.

The B horizon: Otherwise known as the sub soil. This layer is more compact than the topsoil and contains less humus, minerals and organic matter. Due to the presence of clay on this layer, its colour is lighter brown when compared to the top soil. It holds more water than the topsoil.

The C horizon: This zone contains the parent rock which is partly weathered. It has no organic matter and the geological material in this zone is cemented.

The R horizon: This is the bedrock. Contains unweathered parent rock. It is highly compacted. You can find rocks such as granite, basalt and limestone here.

Soil Classification

Soils are classified into groups based on their physical and chemical properties. Classification helps in land use planning and agricultural recommendations.

Common Soil Classification Systems:

- USDA Soil Taxonomy (used in the USA)
- FAO/World Reference Base (WRB) (used internationally)

Examples of Soil Types:

- **Alfisols:** Moderately leached soils found in semi-arid to moist areas
- **Oxisols:** Highly weathered tropical soils with low fertility
- **Ultisols:** Strongly leached, acidic soils, often found in humid climates
- **Entisols:** Young soils with little profile development.

Soil Fertility

Soil fertility is the ability of soil to provide essential nutrients in correct amounts for plant growth. Plant nutrition simply refers to the way which plants take up essential nutrients required for their growth and development from their surroundings. It involves a study of the elements necessary for plant growth.

These essential nutrients are divided into the:

- Macronutrients: which are required in large quantities.
- Micronutrients: which are required in smaller quantities.

Macronutrients include; Nitrogen Phosphorus Potassium, Calcium Magnesium and Sulphur.

Micronutrients include; Iron Manganese, Zinc, Copper, Boron, Molybdenum, Chlorine, Nickel and Cobalt.

Nitrogen, Phosphorus and Potassium forms the key nutrient required by plant.

- Nitrogen: This is required for the overall growth of the plant and also the production of chlorophyll and protein synthesis.
- Phosphorus: This is required for seed germination, root development and energy transfer in plants.
- Potassium: This is required for overall plant health, in water regulation and helps the plant build resistance to diseases.

Nutrient Sources

- Natural: Organic matter and mineral weathering
- Fertilizers: Chemical (e.g., NPK) and organic (e.g., compost)

Nutrient Management Techniques:

1. Soil Testing
2. Fertilizer Application
3. Organic Matter Addition
4. Crop Rotation
5. Lime Application

Common Nutrient Deficiencies:

- Nitrogen: Yellowing of older leaves
- Phosphorus: Stunted growth and purplish leaves
- Potassium: Browning at leaf margins

Soil Organic matter

This is the fraction of the soil that contains plant and animal remains at different stages of decomposition. It can also consist of microbial biomass and products of microbial synthesis. Soil organic matter originates from living organisms, mostly plant tissue. The plant residue contains 60 – 90% moisture while the remaining dry matter consists of carbon, oxygen, hydrogen and small amounts of Sulphur, Nitrogen, Phosphorus, Potassium, Calcium and Magnesium. These are important nutrients from the soil.

Soil organic matter plays a key role in maintaining soil health and fertility.

1. It improves the structure of the soil by binding soil particles together to form a stable and well-drained soil.
2. It acts as a reservoir for plant nutrients and making them available to plants.
3. It provides food and habitat for beneficial soil organisms.
4. It improves the soil's ability to retain more water and keep them available for plants.
5. It helps in carbon sequestration by trapping carbon in the soil and preventing its impact on climate change
6. Surface mulching with coarse organic matter lowers the impact of erosion.

Sources of soil organic matter

Plant residue

1. **Aboveground plant part:** Includes the leaves, stems, branches, and other plant parts that could fall to the ground for decomposition.
2. **Belowground plant part:** Includes the roots, which contribute greatly to organic matter as they decompose.
3. **Green Manure:** Plants grown specifically for incorporation into the soil.

Animal Residues:

1. **Animal Manure:** Decomposed animal waste (dungs)
2. **Animal Remains:** Decomposed dead bodies of animals.

Decomposed Organic Matter:

1. **Compost:** Decomposed organic materials like leaves, grass clippings and food scraps. This is mostly man-made in most cases.
2. **Humus:** very stable, dark-colored, and complex organic matter that results from the advanced decomposition of plant and animal residues.

Other Sources:

1. **Mulch:** Organic materials used to cover the soil surface, such as straw, wood chips, or leaves.
2. **Microbial Biomass:** The living and dead cells of microorganisms in the soil.

Soil Conservation and Erosion Control

What is Soil Conservation?

Soil conservation refers to the protection, management, and improvement of soil to prevent its degradation. It ensures that the soil remains productive and sustainable for agriculture, forestry, construction, and environmental balance.

What is Soil Erosion?

Soil erosion is the removal and transportation of the topsoil layer by natural agents like water, wind, or human activities. The topsoil is the most fertile part, rich in nutrients and organic matter needed for plant growth.

Causes of Soil Erosion

1. **Natural Causes:**
 - Heavy rainfall
 - Strong winds
 - Slope steepness (topography)
 - Lack of vegetation cover

2. Human-Induced Causes:

- Deforestation
- Overgrazing by livestock
- Poor agricultural practices (e.g., continuous cropping, no cover crops)
- Construction and mining without erosion control
- Urbanization

Types of Soil Erosion

A. Water Erosion

1. Splash Erosion:

Raindrops hit bare soil, dislodging particles.

2. Sheet Erosion:

Thin, uniform layers of soil are removed across a wide area.

3. Rill Erosion:

Small channels form on the soil surface due to running water.

4. Gully Erosion:

Deep channels or ditches are formed, often too large to be removed by regular tillage.

B. Wind Erosion

Occurs mostly in dry, sandy, or bare areas where wind can easily blow soil particles away.

C. Other Forms

1. Mass Movement (Landslides):

Movement of soil down a slope due to gravity and saturation.

2. Streambank and Coastal Erosion:

Occurs along riverbanks and shorelines due to flowing water.

Effects of Soil Erosion

1. Loss of fertile topsoil
2. Reduced agricultural productivity
3. Water pollution from sediment runoff
4. Silting of rivers and dams

5. Increased risk of flooding
6. Desertification in severe cases
7. Damage to roads and infrastructure

Soil Conservation Practices

A. Vegetative Measures

1. **Cover Cropping:**
Growing plants like legumes or grasses to protect soil between main crops.
2. **Strip Cropping:**
Alternating rows of crops with grass or legumes to reduce runoff.
3. **Agroforestry:**
Integrating trees with crops to protect against wind and rain.

B. Mechanical Measures

1. **Contour Plowing:**
Plowing along the natural contours of the land to reduce water flow.
2. **Terracing:**
Creating step-like levels on slopes to slow down water and prevent erosion.
3. **Check Dams and Bunds:**
Small barriers built to reduce the speed of runoff water and trap sediments.
4. **Windbreaks or Shelterbelts:**
Rows of trees or shrubs planted to reduce wind speed and trap soil particles.

Soil Management Practices

1. **Mulching:**
Applying a layer of plant residue or synthetic material on soil to protect against erosion and retain moisture.
2. **Conservation Tillage:**
Minimizing soil disturbance to preserve organic matter and reduce erosion.
3. **Crop Rotation:**
Changing crops seasonally to prevent depletion of the same nutrients and reduce pest/disease buildup.

Importance of Soil Conservation

1. Maintains soil fertility and crop yield.
2. Prevents environmental degradation.
3. Supports biodiversity.
4. Ensures long-term sustainability of agriculture.
5. Reduces sedimentation of rivers and water bodies.
6. Prevents land abandonment due to low productivity

Conclusion

Soil science is vital in understanding how soils form, function, and support life. Proper management ensures sustainable agriculture and environmental protection.